

BIJAY SHAKYA

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EDUCATION

Ph.D. in Computer Science

Dakota State University, Madison, SD

Research Areas: Deep Learning, Computer Vision, Federated Learning

August 2023 – December 2026 (Expected)

Master of Science in Computer Science

Dakota State University, Madison, SD

January 2019 – December 2019

Bachelor of Engineering in Computer Engineering

Kathmandu University, Nepal

August 2011 – June 2015

RESEARCH & ACADEMIC EXPERIENCE

Graduate Research Assistant – Dakota State University, Madison, SD

August 2024 – Present

- Currently developing a resource-efficient, multi-level slicing inference framework that incorporates object-density awareness to enhance small-object detection efficiency.
- Engineering a secure federated AI-driven crop anomaly detection framework to detect the early-mid stage crop defects, leveraging hyperspectral imaging (HSI) data for precision agriculture.
- Authored an IEEE Access journal paper benchmarking lightweight You Only Look Once (YOLO) models for real-time small-object detection across diverse datasets and hardware platforms.
- Built end-to-end deep-learning pipeline integrating Visual Language Models (VLMs), object detection, and super-resolution techniques for automated post-disaster damage assessment.
- Graduate teaching assistant for a graduate-level Research and Design Methods course, assisting with course materials preparation, research paper reviews, and assignment grading.

Student Researcher – Dakota State University, Madison, SD

August 2023 – July 2024

- Developed an ensemble deep learning system integrating YOLO model variants using Weighted Boxes Fusion, enhancing detection accuracy by 5% for small objects.
- Researched quantum computing security implications for Smart Grid systems, analyzing potential threats with corresponding mitigation strategies.

Research Assistant – Dakota State University

January 2019 – December 2019

- Delivered technical training workshops on cloud computing (AWS EC2, S3) to 20+ participants with comprehensive training materials and hands-on exercises.
- Co-ordinated undergraduate students in Internet of Things security research work during the Research Experience for Undergraduates (REU) Summer program.
- Co-authored peer-reviewed conference paper on data security frameworks presented at Hawaii International Conference on System Sciences (HICSS) 2021.
- Designed a web application using Laravel, MySQL, and AWS APIs to provide virtual computing lab services for university students.

PROFESSIONAL EXPERIENCE

Application Developer – Ferrandino & Son Inc, Audubon, PA

April 2021 – December 2022

- Engineered and deployed an enterprise application using C #, Blazor, and SQL Server, facilitating contractor assignments across construction business units.
- Leveraged asynchronous REST APIs in C # using HTTPClient and Entity Framework Core, improving performance by 40% and enabling seamless third-party integrations.
- Engineered SQL triggers and assembly scripts to generate executable files for data manipulation operations.

Software Engineer – IMCS Group, Irving, TX

February 2020 – March 2021

- Architected a retail management system handling customer delivery requests and product delivery cost management for FootLocker stores across the USA, Canada, and Europe markets.
- Integrated a web portal that allows authenticated users to update the store's longitude and latitude, which reduced

the manual work by 80%.

- Created an automated reporting system in C #, leveraging ITextSharp and Kendo, reducing 10+ hours of weekly manual reporting.
- Designed and optimized production database schema handling daily transactions by creating indexes, stored procedures, views, and triggers.
- Collaborated with cross-functional team using agile methodologies, participating in sprint planning, and continuous delivery practices to ensure timely product releases.

Software Engineer – Rigo Technologies, Kathmandu, Nepal

October 2017 – December 2018

- Led a backend engineering team of 5 developers building a tenant-based e-learning platform using C #, .NET Core 2.1, PostgreSQL, and Entity Framework Core.
- Launched live production features, including real-time webhook integrations for video analytics, implementing error handling and monitoring to ensure system reliability.
- Mentored new hires and interns, providing technical guidance, and was involved in performance reviews.

Mid-Level Software Engineer – Spiralogics Inc, Kathmandu, Nepal

April 2016 – September 2017

- Collaborated with internal teams to design and develop a workers' compensation and healthcare solution that connects injured parties with healthcare providers for medications and manages billing reports.
- Designed ETL automation module with data validation and email notification that minimized the workload by 80%.
- Optimized database performance by using stored procedures, views, and functions for complex SQL queries.
- Revised the manual SQL queries using Entity Framework, LINQ to simplify CRUD operations, reducing development time and cost.

Software Engineer Intern – Verisk Health, Kathmandu, Nepal

September 2015 – April 2016

- Constructed a Healthcare Report Generating Parser using C#, XML, that expedited document parsing by 25%.
- Implemented ASP.NET MVC to develop healthcare application modules and performed unit testing using N-Unit.
- Involved in the maintenance and features enhancement of the Verisk Health application developed in Web Forms.
- Developed the automation modules to test the application using Selenium.

TECHNICAL SKILLS

- **Programming Languages/Frameworks:** C # (6+ years), Python (2+ years), C, OOP, Javascript, jQuery, HTML5, CSS3, Bootstrap, DevExpress, Kendo UI, Angular 4.
- **ML/AI & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Computer Vision, Object Detection, VLMs, Super-Resolution, Transfer Learning, Federated Learning (Flower 1.25.0).
- **Microsoft .NET Stacks:** Blazor, .NET Core 2.1, ASP.NET MVC, ASP.NET, ADO.NET, Entity Framework Core, REST APIs, LINQ, Web Services.
- **Cloud & DevOps:** AWS (EC2, S3), Unit Testing (NUnit, MSTest, Selenium).
- **Version Control:** Git, GitHub, GitLab, SVN, TFS.
- **Databases:** SQL Server, PostgreSQL, MySQL.
- **Tools & Platforms:** Visual Studio, Swagger, POSTMAN, Cursor, Copilot.

AWARDS & CERTIFICATIONS

- Graduate Research Initiative Award, Dakota State University. August 2023
- Second Place, Third Annual Data Analytics Competition, Dakota State University. December 2024
- **Certifications:**
[AI in Medicine Specialization](#) (DeepLearning.AI) | [Convolutional Neural Networks](#) (DeepLearning.AI) | [Deep Neural Networks with PyTorch](#) (IBM) | [Deep Learning Essentials with Keras](#) (IBM) | [Computer Vision & Image Processing](#) (IBM) | [Hyperparameter Tuning & Optimization](#) (DeepLearning.AI) | [Building a Real-world app with ASP.NET Core and Angular 4+](#) (Udemy)

PUBLICATIONS

- **B. Shakya et al.** Privacy–Utility Tradeoffs in Federated Learning for Hyperspectral Crop Imagery. *International Conference on Applied Computing (CAC)*, *accepted for publication*, 2026.
- **B. Shakya**, C. Hoier, and K. M. Ahmed. [From Pixels to Semantics: A Multi-Stage AI Framework for Structural Damage Detection in Satellite Imagery](#). *arXiv preprint arXiv:2603.22768 (CVPRW)*, 2026.
- **B. Shakya**, O. El-Gayar, J. Kaabi, and K. M. Ahmed. [Small-Object Detection at the Edge: A Pareto-Efficient](#)

- [Benchmark of Lightweight YOLO Models on UAV and Overhead Datasets](#). *IEEE Access*, 2025.
- Y. Wang, K.N. Muthusamy Ragothaman, **B. Shakya**. [Towards Trusted Data Processing for Information and Intelligence Systems](#). *Hawaii International Conference on System Sciences (HICSS)*, 2021.

UNDER REVIEW JOURNALS

- **B. Shakya**, O. El-Gayar, J. Kaabi, and K. M. Ahmed. Adaptive Density-Aware Inference for Efficient Small Object Detection via Multi-Level Slicing. *IEEE Access*, *under review*, 2026.
- **B. Shakya**, P. Vyas, and G. Basyal. A Systematic Review of False Detections in UAV-Based Aerial Imagery: Taxonomy, Causes, and Mitigation Strategies. *IEEE JSTARS*, *under review*, 2026.

TECHNICAL ARTICLES AND RESEARCH OUTREACH

- **TREO Article** — **B. Shakya**, P. Vyas, and G. Basyal. [Mitigation of False Detections in Aerial Based Images \(UAV\): A Systematic Literature Review](#), 2025.
- **TREO Article** — P. Vyas, **B. Shakya**. [Health Misinformation Classification: A Transformer-Based Approach](#), 2024.